Gradient Descent

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Definition 0.1 (Gradient descent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive **step lengths**. **Gradient descent** [1] computes, for $t = 0, 1, 2, \ldots$,

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} - s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

Example 0.1 (Fixed steps on a parabola)**.** For $f(x) = x^2$, $f'(x) = 2x$, so gradient descent with fixed step length s is

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - 2s\, x^{(t)} \\ &= (1 - 2s)\, x^{(t)}, \end{aligned}$$

and $x^{(t)} = (1 - 2s)^t x^{(0)}$.

- With $s = 0.25$, $x^{(t)} = 0.5^t x^{(0)}$, which tends to the minimizer 0.
- With $s = 0.5$, the first step lands on 0, since $1 - 2s = 0$.
- With $s = 1.5$, $x^{(t)} = (-2)^t x^{(0)}$, which moves away from 0 for any $x^{(0)} \neq 0$: too long a step can make gradient descent fail.

Remark (Gradient descent as Newton's method with a simpler Hessian)

Remark 0.1 (Gradient descent as Newton's method with a simpler Hessian). Gradient descent (Definition 0.1) is Newton's method (Definition 5 in Newton's Method) with the Hessian replaced by $\frac{1}{s^{(t)}} \mathbf{I}_p$, so it needs no second derivatives.

Definition 0.2 (Gradient ascent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive step lengths. **Gradient ascent** [1]

repeats, for $t = 0, 1, 2, \ldots$, the update

$$\underset{\sim}{x}^{(t+1)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} + s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

Example 0.2 (Gradient ascent is gradient descent on the negative function)**.** Gradient ascent on f takes the same steps as gradient descent on $-f$, since $\frac{\partial}{\partial \underset{\sim}{x}}(-f) = -\frac{\partial}{\partial \underset{\sim}{x}} f$. In one variable, for $f(x) = -x^2$ the derivative is $f'(x) = -2x$, so gradient ascent with fixed step length s is $x^{(t+1)} = (1 - 2s)\, x^{(t)}$, the iteration of Example 0.1, which tends to the maximizer 0 when $0 < s < 1$.

Definition 0.3 (Line search)**.** In gradient descent (Definition 0.1), a **line search** [2] is a rule that chooses the step length $s^{(t)}$ by examining the values of f along the line $\left\{\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}) : s > 0\right\}$.

Definition 0.4 (Exact line search)**. Exact line search** is the line search (Definition 0.3) that chooses $s^{(t)}$ to minimize $f(\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}))$ over $s > 0$.

Example 0.3 (Exact line search on a parabola)**.** For $f(x) = x^2$ and $x^{(t)} = 2$, the derivative is $f'(2) = 4$, so the line is $\{2 - 4s : s > 0\}$ and the values along it are

$$\begin{aligned} \phi(s) &= f(2 - 4s) \quad \text{(the value at step length } s\text{)} \\ &= (2 - 4s)^2 \quad (f(x) = x^2) \end{aligned}$$

$\phi(s) \geq 0$ for every s, and

$$\begin{aligned} \phi(0.5) &= (2 - 2)^2 \\ &= 0, \end{aligned}$$

so exact line search chooses $s^{(t)} = 0.5$, which lands on the minimizer $x^{(t+1)} = 0$ in one step.

Definition 0.5 (Descent direction)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have a gradient $\underset{\sim}{g} \overset{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$ at $\underset{\sim}{x}$. A direction $\underset{\sim}{d}$ is a **descent direction** for f at $\underset{\sim}{x}$ if $\underset{\sim}{g} \cdot \underset{\sim}{d} < 0$. When $\underset{\sim}{g} \neq \underset{\sim}{0}_{p \times 1}$, $-\underset{\sim}{g}$ is a descent direction, since $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -\|\underset{\sim}{g}\|^2 < 0$.

Example 0.4 (Descent directions for $x_1^2 + x_2^2$ at $(1, 0)^\top$)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ and $\underset{\sim}{x} = (1, 0)^\top$, so

$$\begin{aligned} \underset{\sim}{g} &= (2x_1,\ 2x_2)^\top \\ &= (2, 0)^\top. \end{aligned}$$

- $\underset{\sim}{d} = (-1, 1)^\top$ is a descent direction:

$$\begin{aligned} \underset{\sim}{g} \cdot \underset{\sim}{d} &= -2 + 0 \\ &= -2 \\ &< 0. \end{aligned}$$

- $\underset{\sim}{d} = (0, 1)^\top$ is not: $\underset{\sim}{g} \cdot \underset{\sim}{d} = 0$.
- $-\underset{\sim}{g} = (-2, 0)^\top$ is: $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -4 < 0$.

Theorem 0.1 (A short enough step downhill decreases the function)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$, and let $\underset{\sim}{x} \in \mathbb{R}^p$ be a point with $\underset{\sim}{g} \overset{def}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \neq \underset{\sim}{0}_{p\times 1}$. Then there is an $\bar{s} > 0$ with $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$ for every $s \in (0, \bar{s})$: a short enough step in the descent direction $-\underset{\sim}{g}$ (Definition 0.5) decreases f.*

Proof

Proof. Let $h(s) \overset{\text{def}}{=} f(\underset{\sim}{x} - s\underset{\sim}{g})$. The inner function $s \mapsto \underset{\sim}{x} - s\underset{\sim}{g}$ has derivative $-\underset{\sim}{g}^\top$, so by the vector chain rule

$$\begin{aligned} h'(s) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x} - s\underset{\sim}{g}), && \text{(chain rule)} \\ h'(0) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) && \text{(set } s = 0\text{)} \\ &= -\underset{\sim}{g} \cdot \underset{\sim}{g} && \text{(definition of } \underset{\sim}{g}\text{)} \\ &= -\|\underset{\sim}{g}\|^2 && \text{(squared length)} \\ &< 0. && (\underset{\sim}{g} \neq \underset{\sim}{0}_{p\times 1}) \end{aligned}$$

Since $h'(0) = \lim_{s\to 0}(h(s) - h(0))/s$ is negative, taking $\varepsilon = |h'(0)|$ in the definition of the limit gives an $\bar{s} > 0$ such that $(h(s) - h(0))/s < h'(0) + |h'(0)| = 0$ for $0 < s < \bar{s}$. Multiplying by $s > 0$, $h(s) < h(0)$, that is, $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$. □

Example 0.5 (How short is short enough)**.** In Example 0.1, $f(x) = x^2$ at $x \neq 0$ has $g = 2x$, and $f(x - 2sx) = (1 - 2s)^2x^2 < x^2$ exactly when $|1 - 2s| < 1$, that is, when $0 < s < 1$. So Theorem 0.1 holds with $\bar{s} = 1$, and the step $s = 1.5$ of that example was too long.

Example 0.6 (Gradient descent zigzags)**.** Let $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + b\,x_2^2)$ with $b > 0$, so $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = (x_1,\ bx_2)^\top$ and one gradient step of length s from $\underset{\sim}{x}$ gives $((1-s)x_1,\ (1-bs)x_2)^\top$. For exact line search (Definition 0.4), let $\phi(s) \overset{\text{def}}{=} f$ at that point:

$$\begin{aligned}
\phi(s) &= \tfrac{1}{2}((1-s)^2 x_1^2 + b(1-bs)^2 x_2^2), && \text{(definition of } f\text{)}\\
\phi'(s) &= -(1-s)x_1^2 - b^2(1-bs)x_2^2 && \text{(chain rule, term by term)}\\
&= -x_1^2 + s\,x_1^2 - b^2 x_2^2 + b^3 s\,x_2^2 && \text{(expand the products)}\\
&= s\,(x_1^2 + b^3 x_2^2) - (x_1^2 + b^2 x_2^2). && \text{(collect the terms in } s\text{)}
\end{aligned}$$

For $\underset{\sim}{x} \neq \underset{\sim}{0}_{2\times 1}$, $x_1^2 + b^3 x_2^2 > 0$, so ϕ' is increasing in s with a single zero, and ϕ is minimized at $s^* = \dfrac{x_1^2 + b^2 x_2^2}{x_1^2 + b^3 x_2^2}$. From $\underset{\sim}{x}^{(0)} = (b, 1)^\top$:

$$\begin{aligned}
s^{(0)} &= \frac{b^2 + b^2}{b^2 + b^3} && \text{(substitute } x_1 = b, x_2 = 1\text{)}\\
&= \frac{2}{1+b}, && \text{(cancel } b^2\text{)}
\end{aligned}$$

and, with $r \overset{\text{def}}{=} \frac{b-1}{b+1}$,

$$\begin{aligned}
1 - s^{(0)} &= \frac{1 + b - 2}{1 + b}\\
&= r, && \text{(common denominator)}\\
1 - b\,s^{(0)} &= \frac{1 + b - 2b}{1 + b}\\
&= -r, && \text{(common denominator)}\\
\underset{\sim}{x}^{(1)} &= (r\,b,\ -r \cdot 1)^\top\\
&= r\,(b, -1)^\top. && \text{(the gradient step from } (b,1)^\top\text{)}
\end{aligned}$$

For $b \neq 1$ the same holds at every step, by induction (when $b = 1$, $r = 0$ and $\underset{\sim}{x}^{(1)}$ is already the minimizer): if $\underset{\sim}{x}^{(t)} = (c\,b,\ \pm c)^\top$ for a number $c \neq 0$, then s^*, which depends only on $x_1^2 = c^2 b^2$ and $x_2^2 = c^2$, is again $\frac{2}{1+b}$ (the factor c^2 cancels), and the step multiplies the first entry by r and the second by $-r$. So

$$\begin{aligned}
\underset{\sim}{x}^{(t)} &= (b\,r^t,\ (-r)^t)^\top,\\
f(\underset{\sim}{x}^{(t)}) &= \tfrac{1}{2}(b^2 + b)\,r^{2t}\\
&= r^{2t} f(\underset{\sim}{x}^{(0)}).
\end{aligned}$$

For $b \neq 1$, exactly one entry changes sign at every step: the second when $b > 1$ (then $r > 0$), and the first when $b < 1$ (then $r < 0$). So the iterates zigzag toward the minimizer $\underset{\sim}{0}_{2\times 1}$.

- With $b = 3$, $r = \frac{1}{2}$: $\underset{\sim}{x}^{(1)} = (1.5, -0.5)^\top$ and $\underset{\sim}{x}^{(2)} = (0.75, 0.25)^\top$, and f falls from 6 to 1.5 to 0.375, a quarter at each step.
- With $b = 1$, $r = 0$ and one step reaches the minimizer.
- With $b = 0.01$, $r^2 = (0.99/1.01)^2 \approx 0.96$, so f shrinks by only about 4% per step. Newton's method reaches the minimizer of any such f in one step (Theorem 2 in Newton's Method).

Definition 0.6 (Gradient flow)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives. A **gradient flow** of f is a differentiable curve $\underset{\sim}{x} : [0, \infty) \to \mathbb{R}^p$ that satisfies

$$\frac{d\underset{\sim}{x}(t)}{dt} = -\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}(t)).$$

Example 0.7 (Gradient descent follows the gradient flow)**.** Let $f(x) = \frac{1}{2}x^2$ on $\mathbb{R}$, so $\frac{\partial}{\partial x} f(x) = x$. The gradient flow (Definition 0.6) from $x(0) = x_0$ solves $\frac{dx(t)}{dt} = -x(t)$, whose solution is

$$x(t) = x_0 \exp\{-t\}.$$

Gradient descent with a fixed step length s multiplies x by $1 - s$ at each step, so after k steps it has reached time $t = ks$ with

$$x^{(k)} = x_0 \, (1 - s)^k.$$

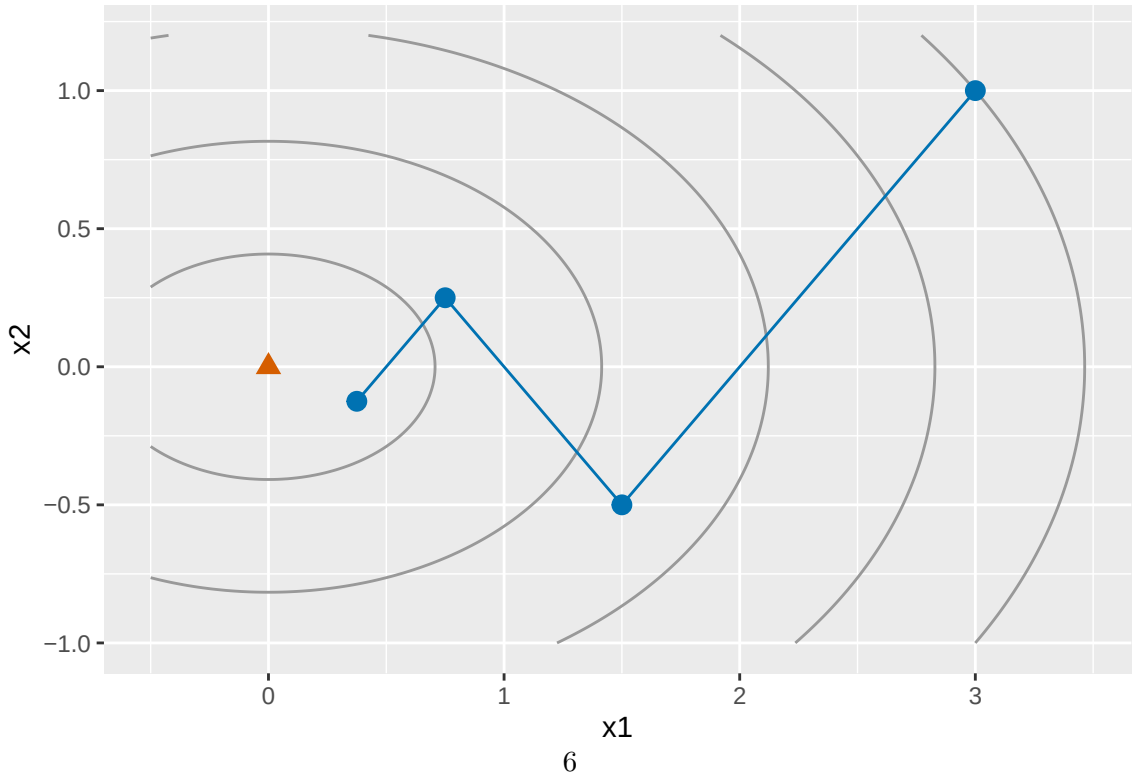
Start at $x_0 = 1$ and run to time $t = 1$:

R

```
x0_gf <- 1
t_gf <- 1
s_gf <- c(0.5, 0.01)
descent_gf <- x0_gf * (1 - s_gf)^(t_gf / s_gf)
flow_gf <- x0_gf * exp(-t_gf)
values_gf <- c(descent_gf, flow = flow_gf)
round(values_gf, 4)
#>                 flow
#> 0.2500 0.3660 0.3679
```

R

```
gz_x1_gz <- seq(-0.5, 3.5, length.out = 201)
gz_x2_gz <- seq(-1, 1.2, length.out = 201)
gz_grid_gz <- expand.grid(x1 = gz_x1_gz, x2 = gz_x2_gz)
gz_grid_gz$f <- 0.5 * (gz_grid_gz$x1^2 + 3 * gz_grid_gz$x2^2)
gz_steps_gz <- 0:3
gz_path_x1_gz <- 3 * 0.5^gz_steps_gz
gz_path_x2_gz <- c(1, -0.5, 0.25, -0.125)
gz_path_gz <- tibble::tibble(x1 = gz_path_x1_gz, x2 = gz_path_x2_gz)
gz_origin_gz <- tibble::tibble(x1 = 0, x2 = 0)
gz_aes_gz <- ggplot2::aes(x1, x2)
gz_aes_z_gz <- ggplot2::aes(z = f)
gz_breaks_gz <- c(0.25, 1, 2.25, 4, 6)
ggplot2::ggplot(gz_grid_gz, gz_aes_gz) +
  ggplot2::geom_contour(
    gz_aes_z_gz, colour = "grey60", breaks = gz_breaks_gz
  ) +
  ggplot2::geom_path(data = gz_path_gz, colour = "#0072B2") +
  ggplot2::geom_point(data = gz_path_gz, colour = "#0072B2", size = 3) +
  ggplot2::geom_point(
    data = gz_origin_gz, colour = "#D55E00", size = 3, shape = 17
  ) +
  ggplot2::labs(x = "x1", y = "x2")
```



Python

```
import matplotlib.pyplot as plt
import numpy as np

gz_x1_gz, gz_x2_gz = np.meshgrid(np.linspace(-0.5, 3.5, 201),
```

Python

```
import numpy as np

x0_gf, t_gf = 1.0, 1.0
s_gf = np.array([0.5, 0.01])
descent_gf = x0_gf * (1 - s_gf) ** (t_gf / s_gf)
flow_gf = x0_gf * np.exp(-t_gf)
print(np.round(np.append(descent_gf, flow_gf), 4));
#> [0.25   0.366  0.3679]
```

With step length $s = 0.5$ the descent reaches 0.25, and with $s = 0.01$ it reaches 0.366, against 0.3679 for the flow. The smaller the step length, the closer the descent stays to the flow.

Remark (Gradient descent is Euler's method for the gradient flow)

Remark 0.2 (Gradient descent is Euler's method for the gradient flow). Gradient descent with a fixed step length s updates $\underset{\sim}{x}^{(t+1)} = \underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)})$. This is Euler's method with time step s for the gradient flow equation.
For a linear function f, the velocity of the flow is the negative normal vector.
By the level sets of a linear function, that vector is orthogonal to every level set, so the flow is too.

Definition 0.7 (Relative change stopping criterion)**.** Let f and $\underset{\sim}{x}^{(0)}, \underset{\sim}{x}^{(1)}, \ldots$ be as in gradient descent, and let $\epsilon > 0$. The **relative change stopping criterion** with tolerance ϵ stops the iteration at the first t for which $f(\underset{\sim}{x}^{(t)}) \neq 0$ and

$$\frac{\left|f(\underset{\sim}{x}^{(t+1)}) - f(\underset{\sim}{x}^{(t)})\right|}{\left|f(\underset{\sim}{x}^{(t)})\right|} < \epsilon.$$

Example 0.8 (Stopping a descent on $f(x) = x^2 + 1$)**.** Let $f(x) = x^2 + 1$. Run gradient descent on f from $x^{(0)} = 2$ with step length $s = 0.25$, and stop at tolerance $\epsilon = 0.01$. Each step maps x to $x - 0.25 \cdot 2x = x/2$:

R

```r
x_rc <- 2
tol_rc <- 0.01
for (t_rc in 0:100) {
  x_next_rc <- x_rc - 0.25 * 2 * x_rc
  cost_rc <- x_rc^2 + 1
  cost_next_rc <- x_next_rc^2 + 1
  diff_rc <- cost_next_rc - cost_rc
  change_rc <- abs(diff_rc) / cost_rc
  below_tol_rc <- change_rc < tol_rc
  if (below_tol_rc) break
  x_rc <- x_next_rc
}
c(t = t_rc, change = change_rc)
#>           t      change
#> 5.000000000 0.002918288
```

Python

```python
f_rc = lambda x: x**2 + 1
x_rc, tol_rc = 2.0, 0.01
for t_rc in range(101):
    x_next_rc = x_rc - 0.25 * 2 * x_rc
    change_rc = abs(f_rc(x_next_rc) - f_rc(x_rc)) / abs(f_rc(x_rc))
    if change_rc < tol_rc:
        break
    x_rc = x_next_rc
print(t_rc, change_rc);
#> 5 0.0029182879377431907
```

The criterion first holds at $t = 5$, where the relative change is 0.00292.

Definition 0.8 (Rosenbrock function)**.** The **Rosenbrock function** on $\mathbb{R}^2$ is

$$f(\underset{\sim}{x}) \stackrel{\text{def}}{=} 100(x_2 - x_1^2)^2 + (1 - x_1)^2, \qquad \underset{\sim}{x} = (x_1, x_2)^\top.$$

It is a standard test problem for optimizers [3].

Example 0.9 (Values of the Rosenbrock function)**.** We evaluate the Rosenbrock function at the minimizer $(1, 1)$, at the origin, and at the standard starting point $(-1.2, 1)$. Then we evaluate it along the curve $x_2 = x_1^2$, which is the floor of a narrow, curved valley.

R

```
rosenbrock_rf <- function(x1, x2) 100 * (x2 - x1^2)^2 + (1 - x1)^2

points_rf <- list(minimizer = c(1, 1), origin = c(0, 0), start = c(-1.2, 1))
sapply(points_rf, function(p) round(rosenbrock_rf(p[1], p[2]), 3))
#> minimizer    origin     start
#>       0.0       1.0      24.2

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf <- c(-1, 0, 0.5, 1, 2)
rbind(
  on_curve = rosenbrock_rf(x_curve_rf, x_curve_rf^2),
  formula = (1 - x_curve_rf)^2
)
#>          [,1] [,2] [,3] [,4] [,5]
#> on_curve    4    1 0.25    0    1
#> formula     4    1 0.25    0    1
```

Python

```
import numpy as np

def rosenbrock_rf(x1, x2):
    return 100 * (x2 - x1 ** 2) ** 2 + (1 - x1) ** 2

points_rf = {"minimizer": (1, 1), "origin": (0, 0), "start": (-1.2, 1)}
print({name: round(rosenbrock_rf(*p), 3) for name, p in points_rf.items()})
#> {'minimizer': 0, 'origin': 1, 'start': 24.2}

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf = np.array([-1.0, 0.0, 0.5, 1.0, 2.0])
print("on_curve:", rosenbrock_rf(x_curve_rf, x_curve_rf ** 2))
#> on_curve: [4.   1.   0.25 0.   1.  ]
print("formula: ", (1 - x_curve_rf) ** 2)
#> formula:  [4.   1.   0.25 0.   1.  ]
```

The function is 0 at $(1, 1)$ and positive everywhere else, because both of its parts are non-negative and vanish together only when $x_1 = 1$ and $x_2 = 1$. So $(1, 1)$ is the global minimizer. At the starting point the value is 24.2, mostly from the first term, because $(-1.2, 1)$ is off the curve $x_2 = x_1^2$. On the curve the function is $(1 - x_1)^2$, which varies slowly compared with the steep walls.

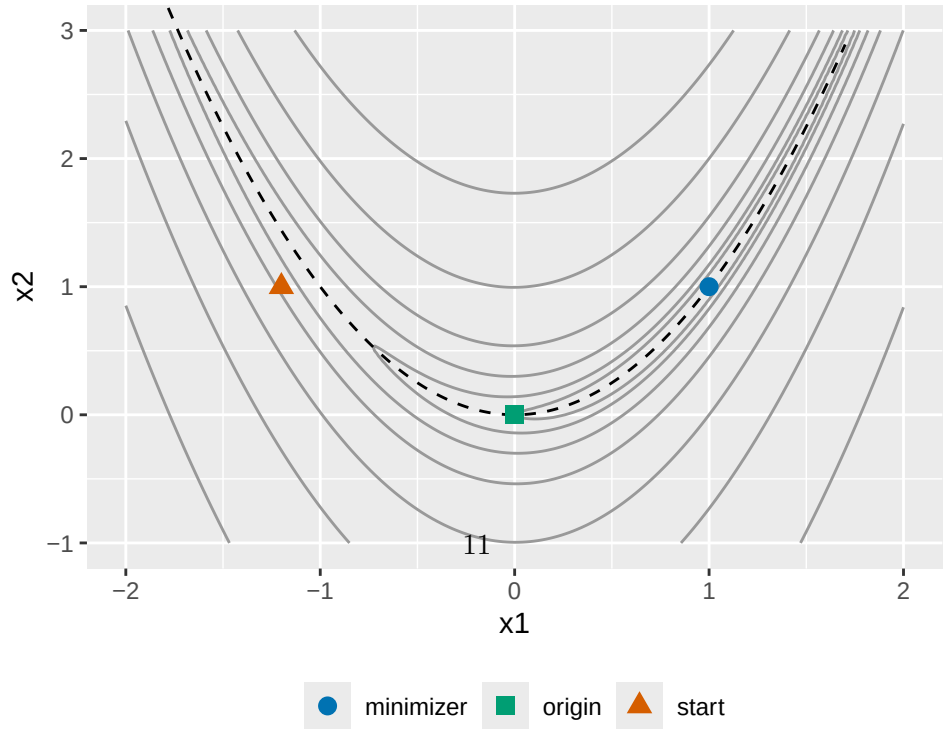
Definition 0.9 (Basin of attraction)**.** For given step lengths, the **basin of attraction** of a local minimizer $\underset{\sim}{x}^*$ under gradient descent is the set of starting points $\underset{\sim}{x}^{(0)}$ from which the iterates converge to $\underset{\sim}{x}^*$ [4].

Example 0.10 (Basins of a double-well function)**.** Let $f(x) = x^4 - 2x^2$, so $\frac{\partial}{\partial x} f(x) = 4x(x-1)(x+1)$ and the stationary points are -1, 0 and 1. Both -1 and 1 are local minimizers, with $f(\pm 1) = -1$, and 0 is a local maximizer. We run gradient descent with the fixed step length $s = 0.05$ for 200 steps from the starting points -1.5, -0.5, -0.1, 0.1, 0.5, 1.5 and 3, and for three steps from $x^{(0)} = 4$.

R

R

```
library(ggplot2)
rf_grid <- expand.grid(
  x1 = seq(-2, 2, length.out = 201),
  x2 = seq(-1, 3, length.out = 201)
)
rf_grid$f <- rosenbrock_rf(rf_grid$x1, rf_grid$x2)
rf_points <- data.frame(
  name = names(points_rf),
  x1 = sapply(points_rf, `[`, 1),
  x2 = sapply(points_rf, `[`, 2)
)
rf_curve <- data.frame(x1 = seq(-2, 1.7, length.out = 200))
rf_curve$x2 <- rf_curve$x1^2
ggplot(rf_grid, aes(x1, x2)) +
  geom_contour(
    aes(z = f),
    colour = "grey60",
    breaks = c(1, 3, 10, 30, 100, 300, 1000)
  ) +
  geom_line(data = rf_curve, linetype = "dashed") +
  geom_point(aes(colour = name, shape = name), data = rf_points, size = 3) +
  scale_colour_manual(
    values = c(minimizer = "#0072B2", origin = "#009E73", start = "#D55E00")
  ) +
  scale_shape_manual(values = c(minimizer = 16, origin = 15, start = 17)) +
  coord_cartesian(xlim = c(-2, 2), ylim = c(-1, 3)) +
  labs(x = "x1", y = "x2", colour = NULL, shape = NULL) +
  theme(legend.position = "bottom")
```



```r
basin_step_bs <- 0.05
basin_starts_bs <- c(-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3)

# 200 steps from every starting point at once.
basin_final_bs <- basin_starts_bs
for (i in 1:200) {
  basin_slope_bs <- 4 * basin_final_bs^3 - 4 * basin_final_bs
  basin_final_bs <- basin_final_bs - basin_step_bs * basin_slope_bs
}
basin_end_bs <- signif(basin_final_bs, 3)
tibble::tibble(start = basin_starts_bs, end = basin_end_bs)
#> # A tibble: 7 x 2
#>   start   end
#>   <dbl> <dbl>
#> 1  -1.5    -1
#> 2  -0.5    -1
#> 3  -0.1    -1
#> 4   0.1     1
#> 5   0.5     1
#> 6   1.5     1
#> 7   3      -1

# The first step from 3.
basin_from_three_bs <- 3 - basin_step_bs * (4 * 3^3 - 4 * 3)

# Three steps from 4: the iterates grow in size.
basin_far_bs <- numeric(4)
basin_far_bs[1] <- 4
for (i in 1:3) {
  basin_slope_bs <- 4 * basin_far_bs[i]^3 - 4 * basin_far_bs[i]
  basin_far_bs[i + 1] <- basin_far_bs[i] - basin_step_bs * basin_slope_bs
}
signif(basin_far_bs, 3)
#> [1]       4.0      -8.0      92.8 -160000.0
```

Python

```
basin_step_bs = 0.05

def basin_next_bs(x):
    return x - basin_step_bs * (4 * x ** 3 - 4 * x)

def basin_gd_bs(x0, n_steps=200):
    x = x0
    for _ in range(n_steps):
        x = basin_next_bs(x)
    return x

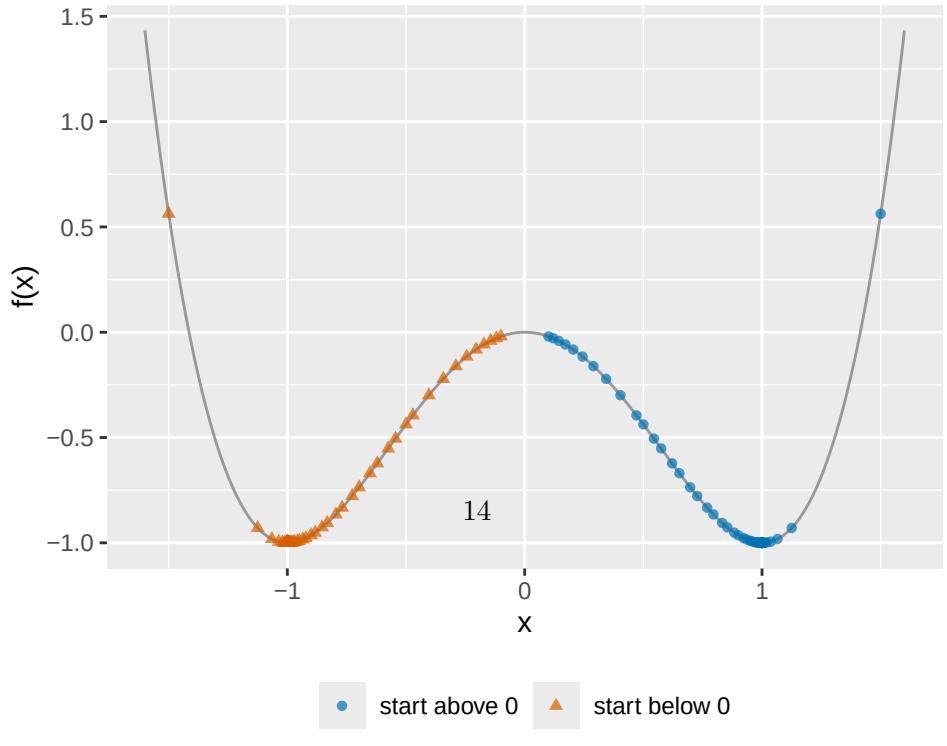
basin_starts_bs = [-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3]
for start in basin_starts_bs:
    print(start, float(f"{basin_gd_bs(start):.3g}"))
#> -1.5 -1.0
#> -0.5 -1.0
#> -0.1 -1.0
#> 0.1 1.0
#> 0.5 1.0
#> 1.5 1.0
#> 3 -1.0

# Three steps from 4: the iterates grow in size.
basin_far_bs = [4.0]
for _ in range(3):
    basin_far_bs.append(basin_next_bs(basin_far_bs[-1]))
print([float(f"{v:.3g}") for v in basin_far_bs])
#> [4.0, -8.0, 92.8, -160000.0]
```

Each of the six starts between -1.5 and 1.5 ends at -1 or 1, whichever lies on its own side of 0, so for these starts the sign of the starting point decides which of the two the run reaches. A run from 0 itself never moves, because $\frac{\partial}{\partial x} f(0) = 0$. The basin of attraction of 1 is not all of $(0, \infty)$: the first step from $x^{(0)} = 3$ lands at -1.8, on the other side of 0, and the run ends at -1. From $x^{(0)} = 4$ the iterates, starting with $x^{(0)}$ itself, are 4, -8, 92.8 and -1.6×10^5, which grow in absolute value at every step shown. Both of these starts are far from 0, where the derivative is large and so the step $s \frac{\partial}{\partial x} f(x)$ is long. From $x^{(0)} = 3$ that step is $96\,s$ long, so the first step stays on the positive side of 0 only when s is below 0.03125. So the basin of 1 depends on s, and with $s = 0.05$ the basin is not a half-line.

R

```
# Uses basin_step_bs and basin_starts_bs from the example above.
basin_runs_bf <- matrix(0, nrow = 31, ncol = 6)
basin_runs_bf[1, ] <- basin_starts_bs[1:6]
for (i in 1:30) {
  basin_prev_bf <- basin_runs_bf[i, ]
  basin_slope_bf <- 4 * basin_prev_bf^3 - 4 * basin_prev_bf
  basin_runs_bf[i + 1, ] <- basin_prev_bf - basin_step_bs * basin_slope_bf
}
basin_side_bf <- ifelse(
  basin_starts_bs[1:6] > 0, "start above 0", "start below 0"
)
basin_x_bf <- as.vector(basin_runs_bf)
basin_side_all_bf <- rep(basin_side_bf, each = 31)
basin_paths_bf <- tibble::tibble(x = basin_x_bf, side = basin_side_all_bf)
basin_paths_bf$cost <- basin_paths_bf$x^4 - 2 * basin_paths_bf$x^2
basin_grid_bf <- seq(-1.6, 1.6, length.out = 321)
basin_curve_bf <- tibble::tibble(
  x = basin_grid_bf, cost = basin_grid_bf^4 - 2 * basin_grid_bf^2
)
basin_colours_bf <- c("start above 0" = "#0072B2", "start below 0" = "#D55E00")
ggplot2::ggplot(basin_curve_bf, ggplot2::aes(x, cost)) +
  ggplot2::geom_line(colour = "grey60") +
  ggplot2::geom_point(
    data = basin_paths_bf,
    ggplot2::aes(colour = side, shape = side), alpha = 0.7
  ) +
  ggplot2::scale_colour_manual(values = basin_colours_bf) +
  ggplot2::labs(x = "x", y = "f(x)", colour = NULL, shape = NULL) +
  ggplot2::theme(legend.position = "bottom")
```



Proposition 0.1 (Divergence of gradient descent on the double-well function)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. If* $x^{(0)}$ *satisfies*

$$x^{(0)^2} > 1 + \frac{1}{2s},$$

then $\left|x^{(t)}\right| \to \infty$ *as* $t \to \infty$*.*

Proof

Proof. The derivative is $\frac{\partial}{\partial x} f(x) = 4x^3 - 4x$, so one step of gradient descent gives

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - s\left(4x^{(t)^3} - 4x^{(t)}\right) && \text{(gradient descent step)} \\ &= -x^{(t)}\left(4sx^{(t)^2} - 4s - 1\right). && \text{(factor out } -x^{(t)}\text{)} \end{aligned}$$

Write $m(x) \stackrel{\text{def}}{=} 4sx^2 - 4s - 1$. Then $\left|x^{(t+1)}\right| = \left|x^{(t)}\right| m(x^{(t)})$ whenever $m(x^{(t)}) \geq 0$. Now $m(x) > 1$ exactly when $x^2 > 1 + 1/(2s)$, and $m(x)$ increases with x^2. So if $x^{(t)}$ satisfies the condition, then $\left|x^{(t+1)}\right| > \left|x^{(t)}\right|$, and $x^{(t+1)}$ satisfies it too. By induction on t, every iterate satisfies it. Since $\left|x^{(t)}\right|$ increases at every step, $x^{(t)^2} \geq x^{(0)^2}$, and so $m(x^{(t)}) \geq m(x^{(0)}) > 1$. Hence $\left|x^{(t)}\right| \geq m(x^{(0)})^t \left|x^{(0)}\right|$, which tends to ∞. □

Corollary 0.1 (Basins of the double-well function are bounded)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. Every start* $x^{(0)}$ *in the basin of attraction of a local minimizer of* f *satisfies*

$$x^{(0)^2} \leq 1 + \frac{1}{2s}.$$

Proof

Proof. The iterates from a start in the basin converge to a local minimizer, so they do not diverge to ∞. By Proposition 0.1, the start therefore does not satisfy $x^{(0)^2} > 1 + 1/(2s)$. □

Example 0.11 (Divergent starts on the double-well function for three step lengths)**.** For $f(x) = x^4 - 2x^2$, Proposition 0.1 says that gradient descent diverges to ∞ from every start with $\left|x^{(0)}\right|$ above $\sqrt{1 + 1/(2s)}$. This bound is 7.14 for $s = 0.01$, 3.32 for $s = 0.05$, and 2.45 for $s = 0.1$. The derivative $4x^3 - 4x$ is unbounded, as is the second derivative $12x^2 - 4$, so the step $s\frac{\partial}{\partial x} f(x)$ grows without bound with x, and a far start overshoots. By Proposition 0.1, every fixed step length fails to converge from some starts. A larger step length lowers the bound, so more starts are guaranteed to diverge. By Corollary 0.1,

each basin of attraction lies within the same bounded interval, which narrows as s grows. In the earlier basin example, $s = 0.05$, and the start 4 lies above the bound 3.32, which is why its iterates grow.

Definition 0.10 (Plateau)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a function with a gradient at every point of a region, and let $\epsilon > 0$ be a tolerance. The region is a **plateau** of f if the Euclidean norm of the gradient satisfies

$$0 < \left\| \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \right\| < \epsilon$$

at every point $\underset{\sim}{x}$ of the region. The function is then nearly flat on the region, and the region contains no stationary point.

Example 0.12 (A cost with no stationary point)**.**
Let $f(x) = \log(1 + \mathrm{e}^{-x})$. Its derivative is $\frac{\partial}{\partial x} f(x) = -1/(1 + \mathrm{e}^{x})$, which is negative for every x, so f has no stationary point at all.
At $x = 0$ the gradient has size 0.5. At $x = 10$ it has size 4.54×10^{-5}, which is small but not zero. A step of gradient descent subtracts the gradient times the step length. The gradient is negative, so the iterates keep moving to the right, more and more slowly. With tolerance $\epsilon = 10^{-4}$, the region $x \geq 10$ is a plateau (Definition 0.10). A run that enters this plateau has not converged; the run has only slowed down.

Definition 0.11 (Armijo condition)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x} \in \mathbb{R}^p$, and let $\underset{\sim}{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$. A step length $s > 0$ satisfies the **Armijo condition** [5] with constant $c \in (0, 1)$ at $\underset{\sim}{x}$ if

$$f(\underset{\sim}{x} - s\underset{\sim}{g}) \leq f(\underset{\sim}{x}) - c\, s\, \|\underset{\sim}{g}\|^2.$$

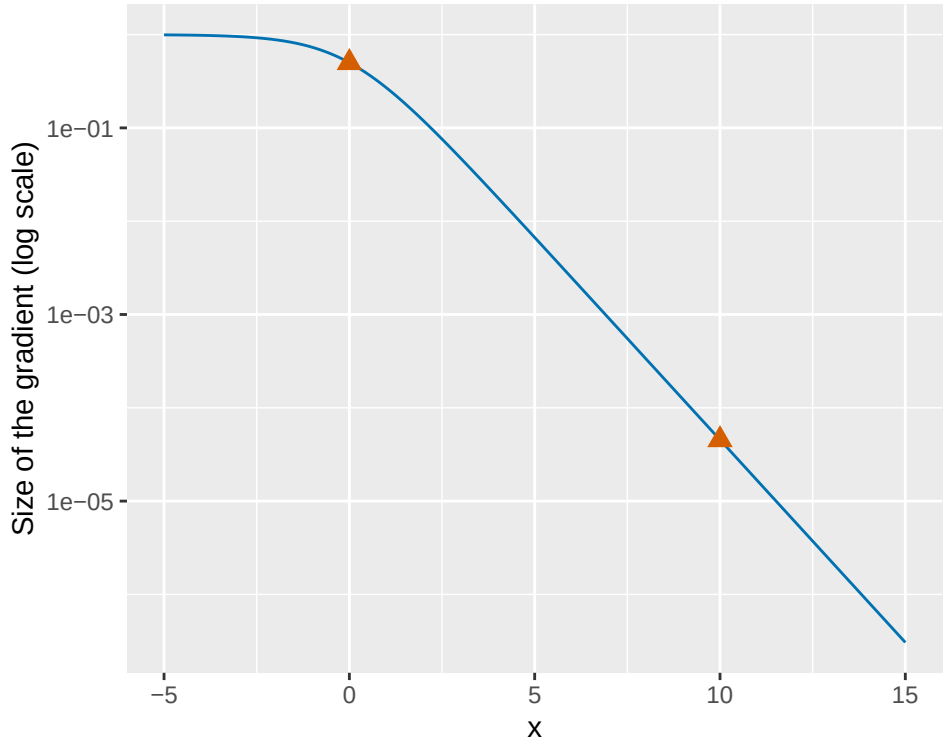
Example 0.13 (The Armijo condition on a parabola)**.** Let $f(x) = x^2$ and $x = 2$, so $g = f'(2) = 4$ and $\|g\|^2 = 16$. With $c = 1/4$, the condition on a step length $s > 0$ is

$$\begin{aligned} (2 - 4s)^2 &\leq 4 - 4s && (f(2 - 4s) \leq f(2) - c\, s\, \|g\|^2) \\ 4 - 16s + 16s^2 &\leq 4 - 4s && (\text{expand the square}) \\ 16s^2 &\leq 12s && (\text{add } 16s - 4 \text{ to both sides}) \\ s &\leq 3/4. && (\text{divide by } 16s > 0) \end{aligned}$$

So $s = 1/2$ satisfies the condition: $f(0) = 0 \leq 4 - 2 = 2$. And $s = 1$ does not: $f(-2) = 4 > 4 - 4 = 0$.

R

```r
plateau_grid_pf <- seq(-5, 15, length.out = 401)
plateau_exp_pf <- exp(plateau_grid_pf)
plateau_size_pf <- 1 / (1 + plateau_exp_pf)
plateau_curve_pf <- tibble::tibble(x = plateau_grid_pf, size = plateau_size_pf)
plateau_marks_x_pf <- c(0, 10)
plateau_marks_exp_pf <- exp(plateau_marks_x_pf)
plateau_marks_size_pf <- 1 / (1 + plateau_marks_exp_pf)
plateau_marks_pf <- tibble::tibble(
  x = plateau_marks_x_pf, size = plateau_marks_size_pf
)
plateau_aes_pf <- ggplot2::aes(x, size)
ggplot2::ggplot(plateau_curve_pf, plateau_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(
    data = plateau_marks_pf, colour = "#D55E00", shape = 17, size = 3
  ) +
  ggplot2::scale_y_log10() +
  ggplot2::labs(x = "x", y = "Size of the gradient (log scale)")
```



Python

```python
import matplotlib.pyplot as plt
import numpy as np

plateau_grid_pf = np.linspace(-5, 15, 401)
plateau_marks_pf = np.array([0, 10])
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(plateau_grid_pf, 1 / (1 + np.exp(plateau_grid_pf)), color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7f6145025e50>]
```

Definition 0.12 (Backtracking line search)**.** Let $s_0 > 0$ be an initial step length, $\beta \in (0,1)$ a shrink factor, and $c \in (0,1)$. **Backtracking line search** [5] is the line search that chooses $s^{(t)}$ to be the largest of $s_0, \beta s_0, \beta^2 s_0, \ldots$ that satisfies the Armijo condition with constant c at $\underset{\sim}{x}^{(t)}$.

Example 0.14 (Backtracking on a parabola)**.** For $f(x) = x^2$ at $x = 2$ with $c = 1/4$, the Armijo condition holds exactly when $s \le 3/4$ (Example 0.13). With $s_0 = 1$ and $\beta = 1/2$, the trial step lengths are $1, 1/2, 1/4, \ldots$. The first, $s = 1$, exceeds $3/4$ and is rejected. The second, $s = 1/2$, is at most $3/4$ and is accepted. So backtracking chooses $s^{(0)} = 1/2$, after two evaluations of f.

Theorem 0.2 (A bound on the error of gradient descent on a quadratic)**.** *Let $\mathbf{S}$ be a $p \times p$ positive definite matrix with eigenvalues $\lambda_1, \ldots, \lambda_p$, let $\underset{\sim}{c} \in \mathbb{R}^p$, and let $f(\underset{\sim}{x}) = \frac{1}{2}\, \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} - \underset{\sim}{c} \cdot \underset{\sim}{x}$, whose unique minimizer is $\underset{\sim}{x}^* = \mathbf{S}^{-1} \underset{\sim}{c}$ (Theorem 2 in Newton's Method). Gradient descent with a fixed step length $s > 0$ from any $\underset{\sim}{x}^{(0)}$ has errors satisfying, for $t = 1, 2, \ldots$,*

$$\left\| \underset{\sim}{x}^{(t)} - \underset{\sim}{x}^* \right\| \le \rho(s)^t \left\| \underset{\sim}{x}^{(0)} - \underset{\sim}{x}^* \right\|, \qquad \rho(s) \overset{def}{=} \max_{1 \le i \le p} |1 - s\,\lambda_i|.$$

Proof

Proof. Let $\underset{\sim}{e}^{(t)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*$ be the error at step t. The gradient of f is $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c}$ (as in the proof of Theorem 2 in Newton's Method), and $\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*$, so

$$\begin{aligned}
\underset{\sim}{e}^{(t+1)} &= \underset{\sim}{x}^{(t)} - s\,(\mathbf{S}\underset{\sim}{x}^{(t)} - \underset{\sim}{c}) - \underset{\sim}{x}^* && \text{(definition of gradient descent)} \\
&= \underset{\sim}{x}^{(t)} - s\,\mathbf{S}(\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*) - \underset{\sim}{x}^* && (\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*) \\
&= (\mathbf{I}_p - s\,\mathbf{S})\, \underset{\sim}{e}^{(t)}. && \text{(collect the terms in } \underset{\sim}{e}^{(t)})
\end{aligned}$$

By induction on t, $\underset{\sim}{e}^{(t)} = (\mathbf{I}_p - s\,\mathbf{S})^t\, \underset{\sim}{e}^{(0)}$. By the spectral theorem, write $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$. Then $\mathbf{I}_p - s\,\mathbf{S} = \mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})\mathbf{Q}^\top$ because $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$, and the tth power cancels each inner $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p$, so

$$\begin{aligned}
\left\| \underset{\sim}{e}^{(t)} \right\| &= \left\| \mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})^t \mathbf{Q}^\top \underset{\sim}{e}^{(0)} \right\| && \text{(the factorization above)} \\
&= \left\| (\mathbf{I}_p - s\,\mathbf{\Lambda})^t \mathbf{Q}^\top \underset{\sim}{e}^{(0)} \right\| && \text{(orthogonal matrices preserve length)} \\
&\le \rho(s)^t \left\| \mathbf{Q}^\top \underset{\sim}{e}^{(0)} \right\| && \text{(each coordinate is multiplied by } (1 - s\lambda_i)^t) \\
&= \rho(s)^t \left\| \underset{\sim}{e}^{(0)} \right\|. && \text{(orthogonal matrices preserve length)}
\end{aligned}$$

The length-preserving steps apply orthogonal matrices preserve length. The inequality holds because $\mathbf{I}_p - s\,\mathbf{\Lambda}$ is diagonal, so its tth power multiplies coordinate i by $(1 - s\,\lambda_i)^t$, whose absolute value is at most $\rho(s)^t$. □

Example 0.15 (The error multiplier for a step length of one quarter)**.** Let $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1, and $\underset{\sim}{c} = \underset{\sim}{0}_{2\times 1}$, so $\underset{\sim}{x}^* = \underset{\sim}{0}_{2\times 1}$. With $s = 0.25$, the factors $1 - s\,\lambda_i$ are 0.25 and 0.75, so $\rho(s) = 0.75$ and

$$\left\|\underset{\sim}{x}^{(t)}\right\| \le 0.75^t \left\|\underset{\sim}{x}^{(0)}\right\|.$$

Along the eigenvector for $\lambda = 3$ the error shrinks fast, and along the eigenvector for $\lambda = 1$ it shrinks slowly; the bound follows the slower one.

Theorem 0.3 (The best constant step length)**.** *In the setting of Theorem 0.2, let* $\lambda_{\min}$ *and* $\lambda_{\max}$ *be the smallest and largest eigenvalues of* $\mathbf{S}$*. The factor* $\rho(s) = \max_i |1 - s\,\lambda_i|$ *of Theorem 0.2 is smallest over* $s > 0$ *at*

$$s^* = \frac{2}{\lambda_{\min} + \lambda_{\max}},$$

where

$$\rho(s^*) = \frac{\kappa - 1}{\kappa + 1}, \qquad \kappa = \kappa(\mathbf{S}),$$

the condition number of $\mathbf{S}$*.*

Proof

Proof. For $s > 0$, each $1 - s\,\lambda_i$ lies between $u_{\min} \overset{\text{def}}{=} 1 - s\,\lambda_{\max}$ and $u_{\max} \overset{\text{def}}{=} 1 - s\,\lambda_{\min}$, so $|1 - s\,\lambda_i| \le \max(u_{\max}, -u_{\min})$. Both $\lambda_{\min}$ and $\lambda_{\max}$ occur among the λ_i, and $|u_{\max}| \ge u_{\max}$ and $|u_{\min}| \ge -u_{\min}$, so the maximum is attained:

$$\rho(s) = \max(1 - s\,\lambda_{\min},\ s\,\lambda_{\max} - 1).$$

The first expression is strictly decreasing in s, and the second is strictly increasing, and they coincide when

$$\begin{aligned} 1 - s\,\lambda_{\min} &= s\,\lambda_{\max} - 1 && \text{(set the two expressions equal)} \\ 2 &= s\,(\lambda_{\min} + \lambda_{\max}) && \text{(collect the terms in } s\text{)} \\ s &= \frac{2}{\lambda_{\min} + \lambda_{\max}} = s^*. && \text{(divide)} \end{aligned}$$

For $s < s^*$, $\rho(s)$ is at least the decreasing expression, which exceeds its value at s^*. For $s > s^*$, $\rho(s)$ is at least the increasing expression, which exceeds its value at s^*. So s^* is the minimizer, and

$$\begin{aligned}
\rho(s^*) &= 1 - \frac{2\,\lambda_{\min}}{\lambda_{\min} + \lambda_{\max}} && \text{(first expression at } s^*\text{)} \\
&= \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} && \text{(common denominator)} \\
&= \frac{\kappa - 1}{\kappa + 1}. && \text{(divide the numerator and denominator by } \lambda_{\min}\text{)}
\end{aligned}$$

□

Example 0.16 (The best step length for eigenvalues 3 and 1)**.** For $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1, Theorem 0.3 gives $s^* = 2/(3+1) = 0.5$, the condition number $\kappa = 3$, and $\rho(s^*) = (3-1)/(3+1) = 0.5$. A grid search over $s = 0.01, 0.02, \ldots, 1$ finds the smallest $\rho(s)$ at $s = 0.5$, the same step length. Each step then multiplies the error by at most 0.5.

Exercise 0.1 (Which step lengths shrink the error bound)**.** In the setting of Theorem 0.2, find all $s > 0$ with $\rho(s) < 1$, and write the answer as a condition on s that involves $\lambda_{\max}$.

Solution

Solution 0.1. $\rho(s) < 1$ says that $|1 - s\,\lambda_i| < 1$ for every i, which is $-1 < 1 - s\,\lambda_i < 1$. The right inequality holds for every $s > 0$ because $\lambda_i > 0$. The left inequality is $s\,\lambda_i < 2$, that is, $s < 2/\lambda_i$. These p inequalities, one for each eigenvalue, all hold exactly when the inequality for the largest eigenvalue holds, since it is the most restrictive:

$$0 < s < \frac{2}{\lambda_{\max}}.$$

Exercise 0.2 (The best step length for eigenvalues 1 and 9)**.** In gradient descent, a fixed step length that is too short converges slowly, and one that is too long diverges. Theorem 0.3 gives the best one from the eigenvalues of $\mathbf{S}$.
Let $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$. Using Theorem 0.3, find s^*.

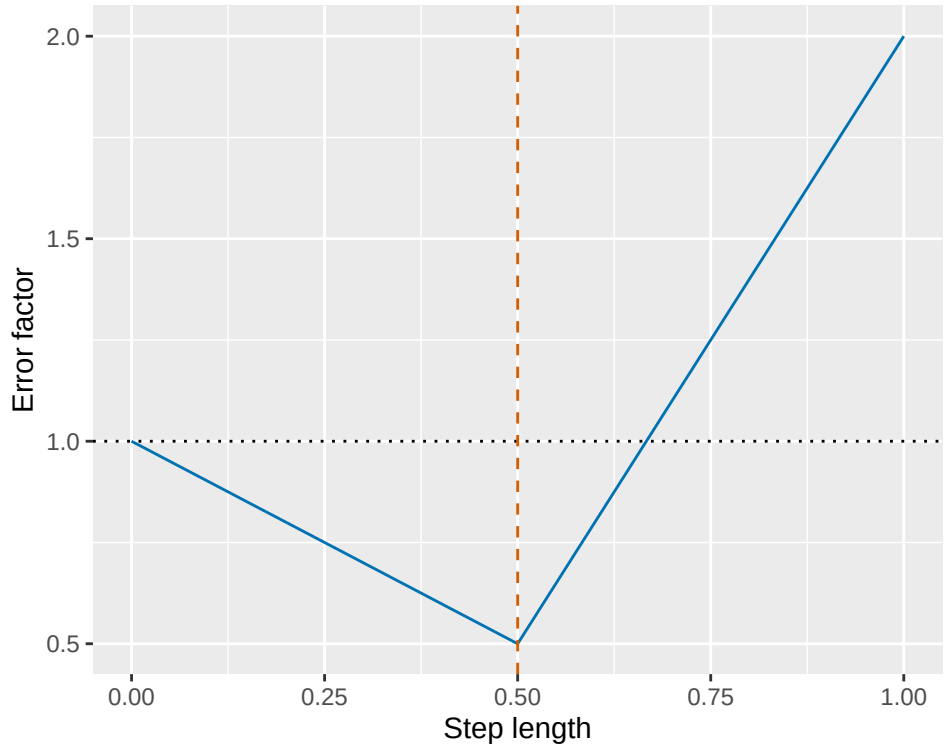
Solution

Solution 0.2. $\mathbf{S}$ is diagonal, so its eigenvalues are its diagonal entries 1 and 9. Then

$$s^* = \frac{2}{1+9} = 0.2.$$

R

```r
bcs_grid_pf <- seq(0, 1, length.out = 401)
bcs_first_pf <- abs(1 - 3 * bcs_grid_pf)
bcs_second_pf <- abs(1 - bcs_grid_pf)
bcs_rho_pf <- pmax(bcs_first_pf, bcs_second_pf)
bcs_curve_pf <- tibble::tibble(s = bcs_grid_pf, rho = bcs_rho_pf)
bcs_aes_pf <- ggplot2::aes(s, rho)
ggplot2::ggplot(bcs_curve_pf, bcs_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_hline(yintercept = 1, linetype = "dotted") +
  ggplot2::geom_vline(
    xintercept = 0.5, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::labs(x = "Step length", y = "Error factor")
```



Python

```python
import matplotlib.pyplot as plt
import numpy as np

bcs_grid_pf = np.linspace(0, 1, 401)
bcs_rho_pf = np.maximum(np.abs(1 - 3 * bcs_grid_pf), np.abs(1 - bcs_grid_pf))
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(bcs_grid_pf, bcs_rho_pf, color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7f614518a990>]
ax.axhline(1, color="black", linestyle=":")
#> <matplotlib.lines.Line2D object at 0x7f614518aad0>
ax.axvline(0.5, color="#D55E00", linestyle="--")
#> <matplotlib.lines.Line2D object at 0x7f614518a210>
ax.set_xlabel("Step length")
```

Exercise 0.3 (The best error factor for eigenvalues 1 and 9)**.** In the setting of Exercise 0.2, find $\rho(s^*)$.

Solution

Solution 0.3. The condition number is $\kappa(\mathbf{S}) = 9/1 = 9$, so by Theorem 0.3

$$\rho(s^*) = \frac{9-1}{9+1} = 0.8.$$

Exercise 0.4 (Steps to cut the error bound by a factor of ten)**.** In the setting of Exercise 0.2 and Exercise 0.3, find the smallest $t = 1, 2, ...$ with $\rho(s^*)^t \leq 0.1$.

Solution

Solution 0.4. From Solution 0.3, $\rho(s^*) = 0.8$. Taking logarithms, $\rho(s^*)^t \leq 0.1$ becomes $t \ln \rho(s^*) \leq \ln 0.1$. Here $\ln \rho(s^*) < 0$, so dividing by it flips the inequality: $t \geq \ln 0.1 / \ln \rho(s^*) \approx 10.32$. The smallest such t is 11.

Definition 0.13 (Momentum coefficient)**.** Let $\underset{\sim}{g}^{(1)}, \underset{\sim}{g}^{(2)}, ...$ be vectors in $\mathbb{R}^p$, and let $\underset{\sim}{v}^{(t)}$ be a running **velocity** built from them. The **momentum coefficient** $\beta \in [0, 1)$ is the factor that multiplies the previous velocity in

$$\underset{\sim}{v}^{(t)} \stackrel{\text{def}}{=} \beta \underset{\sim}{v}^{(t-1)} + \underset{\sim}{g}^{(t)},$$

for $t = 1, 2, ...$, with $\underset{\sim}{v}^{(0)} \stackrel{\text{def}}{=} \underset{\sim}{0}$.

Example 0.17 (How far back the velocity remembers)**.** Unrolling the recursion, the vector $\underset{\sim}{g}^{(t-k)}$ enters $\underset{\sim}{v}^{(t)}$ with factor β^k, for $k < t$. These factors add up to at most $\sum_{k=0}^{\infty} \beta^k = 1/(1-\beta)$. As a rule of thumb, the velocity is therefore dominated by the last $1/(1-\beta)$ vectors.

R

```r
beta_mc <- c(0, 0.5, 0.9, 0.99)
k_mc <- 10
weight_mc <- beta_mc^k_mc
memory_mc <- 1 / (1 - beta_mc)
weight_rounded_mc <- signif(weight_mc, 2)
# nolint start: undesirable_function_linter.
rbind(beta = beta_mc, weight = weight_rounded_mc, memory = memory_mc)
#>        [,1]    [,2]  [,3]   [,4]
#> beta      0 0.50000  0.90   0.99
#> weight    0 0.00098  0.35   0.90
#> memory    1 2.00000 10.00 100.00
# nolint end
```

Python

```python
beta_mc = [0, 0.5, 0.9, 0.99]
k_mc = 10
for b_mc in beta_mc:
    print(b_mc, float(f"{b_mc**k_mc:.2g}"), round(1 / (1 - b_mc)));
#> 0 0.0 1
#> 0.5 0.00098 2
#> 0.9 0.35 10
#> 0.99 0.9 100
```

At $\beta = 0$ the velocity is the current vector alone, and at $\beta = 0.99$ it combines about 100 vectors.

Definition 0.14 (Momentum (heavy-ball method))**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, let $s > 0$ be a step length, and let $\beta \in [0, 1)$ be a momentum coefficient. **Momentum** (the **heavy-ball method**) updates, for $t = 1, 2, \ldots$, by

$$\underset{\sim}{v}^{(t)} \stackrel{\text{def}}{=} \beta\, \underset{\sim}{v}^{(t-1)} + \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t-1)})$$
$$\underset{\sim}{x}^{(t)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t-1)} - s\, \underset{\sim}{v}^{(t)},$$

with $\underset{\sim}{v}^{(0)} \stackrel{\text{def}}{=} \underset{\sim}{0}$.

Example 0.18 (Momentum on a one-dimensional quadratic)**.** Let $f(x) = \frac{1}{2}x^2$, so $\frac{\partial}{\partial x} f(x) = x$. Start at $x^{(0)} = 1$ with $s = 0.1$, and take 20 updates with $\beta = 0$, which is

plain gradient descent, and with $\beta = 0.9$:

R

```
x0_mo <- 1
s_mo <- 0.1
beta_mo <- c(plain = 0, momentum = 0.9)
n_mo <- 20
x_mo <- rep(x0_mo, 2)
v_mo <- c(0, 0)
path_mo <- matrix(NA_real_, nrow = n_mo, ncol = 2)
for (k_mo in seq_len(n_mo)) {
  v_mo <- beta_mo * v_mo + x_mo
  x_mo <- x_mo - s_mo * v_mo
  path_mo[k_mo, ] <- x_mo
}
last_mo <- path_mo[n_mo, ]
lowest_mo <- apply(path_mo, 2, min)
# nolint start: undesirable_function_linter.
summary_mo <- rbind(last = last_mo, lowest = lowest_mo)
# nolint end
round(summary_mo, 4)
#>          [,1]    [,2]
#> last   0.1216  0.3447
#> lowest 0.1216 -0.6037
```

Python

```
s_mo, n_mo = 0.1, 20
for beta_mo in (0.0, 0.9):
    x_mo, v_mo, path_mo = 1.0, 0.0, []
    for _ in range(n_mo):
        v_mo = beta_mo * v_mo + x_mo
        x_mo = x_mo - s_mo * v_mo
        path_mo.append(x_mo)
    print(beta_mo, round(path_mo[-1], 4), round(min(path_mo), 4));
#> 0.0 0.1216 0.1216
#> 0.9 0.3447 -0.6037
```

After 20 updates, plain descent has reached 0.122 and momentum 0.345. Momentum overshoots the minimizer 0, down to -0.604 at update 9, and then swings back. In this

run, with a small step length, the swing leaves momentum behind plain descent after 20 updates.

Definition 0.15 (Adaptive method)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, let $s > 0$ be a step length, let $\rho \in [0, 1)$ be a decay factor, and let $\epsilon > 0$. Write $g_j^{(t)}$ for coordinate j of $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t-1)})$. An **adaptive method** is defined, for $t = 1, 2, \ldots$ and each coordinate $j = 1, \ldots, p$,

$$u_j^{(t)} \stackrel{\text{def}}{=} \rho\, u_j^{(t-1)} + (1 - \rho) \left(g_j^{(t)}\right)^2$$

$$x_j^{(t)} \stackrel{\text{def}}{=} x_j^{(t-1)} - s\, \frac{g_j^{(t)}}{\sqrt{u_j^{(t)}} + \epsilon},$$

with $u_j^{(0)} \stackrel{\text{def}}{=} 0$.

Example 0.19 (Two coordinates on very different scales)**.** Suppose the two coordinates of the gradient start at $g_1^{(1)} = 100$ and $g_2^{(1)} = 0.01$, and both shrink by a factor of 0.5 at each update. Take $\rho = 0.9$, and look at the first 5 updates. Dividing by the root of the running average gives the factor $g_j^{(t)} / \sqrt{u_j^{(t)}}$ that multiplies s in the update of each coordinate:

R

```r
s_am <- c(100, 0.01)
rho_am <- 0.9
q_am <- 0.5
n_am <- 5
k_am <- seq_len(n_am) - 1
g_am <- outer(q_am^k_am, s_am)
u_am <- matrix(0, n_am, 2)
u_prev_am <- c(0, 0)
for (t_am in seq_len(n_am)) {
  u_prev_am <- rho_am * u_prev_am + (1 - rho_am) * g_am[t_am, ]^2
  u_am[t_am, ] <- u_prev_am
}
factor_am <- g_am / sqrt(u_am)
round(factor_am, 3)
#>       [,1]  [,2]
#> [1,] 3.162 3.162
#> [2,] 1.474 1.474
#> [3,] 0.755 0.755
#> [4,] 0.395 0.395
#> [5,] 0.208 0.208
```

Python

```python
import numpy as np

s_am, rho_am, q_am, n_am = [100, 0.01], 0.9, 0.5, 5
g_am = np.outer(q_am ** np.arange(n_am), s_am)
u_am = np.zeros(2)
factor_am = []
for t_am in range(n_am):
    u_am = rho_am * u_am + (1 - rho_am) * g_am[t_am] ** 2
    factor_am.append(g_am[t_am] / np.sqrt(u_am))
print(np.round(factor_am, 3));
#> [[3.162 3.162]
#>  [1.474 1.474]
#>  [0.755 0.755]
#>  [0.395 0.395]
#>  [0.208 0.208]]
```

Plain gradient descent would move the first coordinate 10^4 times as far as the second. The factors are the same for both coordinates, so the scale of a coordinate no longer

matters. They also shrink from 3.16 to 0.21, because the running average still holds the earlier, larger values. Dividing by the current gradient alone would give a factor of 1 at every update.

Remark (Early adaptive methods)

Remark 0.3 (Early adaptive methods). RMSProp is an early example of an adaptive method [6]. AdaGrad, an earlier method, replaces the exponentially weighted average by a plain sum of squared gradients, $u_j^{(t)} = u_j^{(t-1)} + \left(g_j^{(t)}\right)^2$.

Definition 0.16 (Adam)**.** Let f, $\underset{\sim}{x}^{(0)}$, s and ϵ be as in an adaptive method, and let $\beta_1, \beta_2 \in [0, 1)$, where β_2 plays the role of the decay factor ρ. Write $\underset{\sim}{g}^{(t)}$ for $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t-1)})$. The **Adam** method [7] is defined, for $t = 1, 2, \ldots$,

$$
\begin{aligned}
\underset{\sim}{\mu}^{(t)} &\stackrel{\text{def}}{=} \beta_1 \, \underset{\sim}{\mu}^{(t-1)} + (1 - \beta_1) \, \underset{\sim}{g}^{(t)} \\
u_j^{(t)} &\stackrel{\text{def}}{=} \beta_2 \, u_j^{(t-1)} + (1 - \beta_2) \left(g_j^{(t)}\right)^2 \\
\underset{\sim}{\hat{\mu}}^{(t)} &\stackrel{\text{def}}{=} \frac{\underset{\sim}{\mu}^{(t)}}{1 - \beta_1^t} \\
\hat{u}_j^{(t)} &\stackrel{\text{def}}{=} \frac{u_j^{(t)}}{1 - \beta_2^t} \\
x_j^{(t)} &\stackrel{\text{def}}{=} x_j^{(t-1)} - s \, \frac{\hat{m}_j^{(t)}}{\sqrt{\hat{u}_j^{(t)}} + \epsilon},
\end{aligned}
$$

for each coordinate $j = 1, \ldots, p$, with $\underset{\sim}{\mu}^{(0)} \stackrel{\text{def}}{=} \underset{\sim}{0}$ and $u_j^{(0)} \stackrel{\text{def}}{=} 0$.

Example 0.20 (The first Adam step ignores the size of the gradient)**.** Let $f(x) = \frac{1}{2}x^2$, so $\frac{\partial}{\partial x} f(x) = x$, and take $s = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. At $t = 1$ the corrections for the zero start of $\underset{\sim}{\mu}$ and u_j cancel the factors $1 - \beta_1$ and $1 - \beta_2$, so $\hat{m}^{(1)} = g^{(1)}$ and $\hat{u}^{(1)} = \left(g^{(1)}\right)^2$. The first step is then $s \, g^{(1)} / (|g^{(1)}| + \epsilon)$, which is about s in size. Start at $x^{(0)} = 2$ and at $x^{(0)} = 2000$:

R

```r
s_ad <- 0.1
beta1_ad <- 0.9
beta2_ad <- 0.999
eps_ad <- 1e-8
x0_ad <- c(2, 2000)
g_ad <- x0_ad
m_ad <- (1 - beta1_ad) * g_ad
u_ad <- (1 - beta2_ad) * g_ad^2
m_hat_ad <- m_ad / (1 - beta1_ad)
u_hat_ad <- u_ad / (1 - beta2_ad)
step_ad <- s_ad * m_hat_ad / (sqrt(u_hat_ad) + eps_ad)
round(step_ad, 4)
#> [1] 0.1 0.1
```

Python

```python
import numpy as np

s_ad, beta1_ad, beta2_ad, eps_ad = 0.1, 0.9, 0.999, 1e-8
g_ad = np.array([2.0, 2000.0])
m_hat_ad = (1 - beta1_ad) * g_ad / (1 - beta1_ad)
u_hat_ad = (1 - beta2_ad) * g_ad ** 2 / (1 - beta2_ad)
print(np.round(s_ad * m_hat_ad / (np.sqrt(u_hat_ad) + eps_ad), 4));
#> [0.1 0.1]
```

Both first steps have size 0.1, though the gradients differ by a factor of 1000. Plain gradient descent would move $s\,g^{(1)}$, which is 0.2 and 200.

Definition 0.17 (Piecewise constant schedule)**.** Let $s^{(0)} > 0$ be the step length before update 1, let $\gamma \in (0,1)$ be a drop factor, and let $t_1 < t_2 < \cdots$ be update counts. The **piecewise constant schedule** is

$$s^{(t)} \overset{\text{def}}{=} s^{(0)}\,\gamma^{k(t)}, \qquad k(t) \overset{\text{def}}{=} \big|\{j : t_j \le t\}\big|.$$

Example 0.21 (A ten-fold drop twice)**.** Take $s^{(0)} = 0.1$ and $\gamma = 0.1$, with drops at $t_1 = 30$ and $t_2 = 60$. The step length just before and at each drop is:

R

```
s0_pc <- 0.1
decay_pc <- 0.1
drops_pc <- c(30, 60)
t_pc <- c(0, 29, 30, 59, 60, 90)
dropped_pc <- outer(drops_pc, t_pc, "<=")
k_pc <- colSums(dropped_pc)
step_pc <- s0_pc * decay_pc^k_pc
tibble::tibble(t = t_pc, step_length = step_pc)
#> # A tibble: 6 x 2
#>       t step_length
#>   <dbl>       <dbl>
#> 1     0       0.1
#> 2    29       0.1
#> 3    30       0.01
#> 4    59       0.01
#> 5    60       0.001
#> 6    90       0.001
```

Python

```
s0_pc, decay_pc, drops_pc = 0.1, 0.1, [30, 60]
for t_pc in [0, 29, 30, 59, 60, 90]:
    k_pc = sum(d_pc <= t_pc for d_pc in drops_pc)
    print(t_pc, round(s0_pc * decay_pc ** k_pc, 6));
#> 0 0.1
#> 29 0.1
#> 30 0.01
#> 59 0.01
#> 60 0.001
#> 90 0.001
```

The step length stays fixed between drops. It falls by the factor γ exactly at $t = 30$ and $t = 60$.

Definition 0.18 (Linear decay)**.** Let τ be a number of updates, and let $s^{(0)}, s^{(\tau)} > 0$ be the step lengths before update 1 and after update τ. The **linear decay** schedule is

$$s^{(t)} \stackrel{\text{def}}{=} (1 - \zeta)\, s^{(0)} + \zeta\, s^{(\tau)}, \qquad \zeta \stackrel{\text{def}}{=} \min\left(\frac{t}{\tau}, 1\right).$$

Example 0.22 (Linear decay between two step lengths)**.** Take $s^{(0)} = 0.1$, $s^{(\tau)} = 0.001$ and $\tau = 90$. The step length at $t = 0, 45, 90, 120$ is:

R

```
s0_ld <- 0.1
s_end_ld <- 0.001
tau_ld <- 90
t_ld <- c(0, 45, 90, 120)
frac_ld <- pmin(t_ld / tau_ld, 1)
step_ld <- (1 - frac_ld) * s0_ld + frac_ld * s_end_ld
tibble::tibble(t = t_ld, step_length = step_ld)
#> # A tibble: 4 x 2
#>       t step_length
#>   <dbl>       <dbl>
#> 1     0      0.1
#> 2    45      0.0505
#> 3    90      0.001
#> 4   120      0.001
```

Python

```
s0_ld, sT_ld, tau_ld = 0.1, 0.001, 90
for t_ld in [0, 45, 90, 120]:
    frac_ld = min(t_ld / tau_ld, 1)
    print(t_ld, round((1 - frac_ld) * s0_ld + frac_ld * sT_ld, 6));
#> 0 0.1
#> 45 0.0505
#> 90 0.001
#> 120 0.001
```

The step length falls by the same amount at each update. It then stays at 0.001 for $t \geq \tau$.

Definition 0.19 (Exponential decay)**.** Let $s^{(0)} > 0$ be the step length before update 1, and let $\gamma \in (0, 1)$ be a decay factor. The **exponential decay** schedule is

$$s^{(t)} \stackrel{\text{def}}{=} s^{(0)} \gamma^t.$$

Example 0.23 (The same factor over two run lengths)**.** Take $s^{(0)} = 0.1$ and $\gamma = 0.99$. The step length at $t = 1000$ and at $t = 100000$ is:

R

```
s0_ed <- 0.1
decay_ed <- 0.99
t_ed <- c(1000, 100000)
step_ed <- s0_ed * decay_ed^t_ed
tibble::tibble(t = t_ed, step_length = step_ed)
#> # A tibble: 2 x 2
#>        t step_length
#>    <dbl>       <dbl>
#> 1   1000  0.00000432
#> 2 100000  0
```

Python

```
s0_ed, decay_ed = 0.1, 0.99
for t_ed in [1000, 100000]:
    print(t_ed, s0_ed * decay_ed ** t_ed);
#> 1000 4.317124741065786e-06
#> 100000 0.0
```

At $t = 1000$ the step length is about 4.3×10^{-6}. At $t = 100000$ it is 0 in floating-point arithmetic. One factor gives two very different schedules.

Definition 0.20 (Cosine decay)**.** Let τ be a number of updates, and let $s^{(0)} > 0$ be the step length before update 1. The **cosine decay** schedule [8] is

$$s^{(t)} \stackrel{\text{def}}{=} \frac{s^{(0)}}{2}\left(1 + \cos\frac{\pi t}{\tau}\right), \qquad t = 0, 1, \ldots, \tau.$$

Example 0.24 (Cosine decay over a fixed run)**.** Take $s^{(0)} = 0.1$ and $\tau = 90$. The step length at $t = 0, 15, 45, 75, 90$ is:

R

```r
s0_cd <- 0.1
tau_cd <- 90
t_cd <- c(0, 15, 45, 75, 90)
angle_cd <- pi * t_cd / tau_cd
step_cd <- s0_cd / 2 * (1 + cos(angle_cd))
tibble::tibble(t = t_cd, step_length = step_cd)
#> # A tibble: 5 x 2
#>       t step_length
#>   <dbl>       <dbl>
#> 1     0     0.1
#> 2    15     0.0933
#> 3    45     0.05
#> 4    75     0.00670
#> 5    90     0
```

Python

```python
import math

s0_cd, tau_cd = 0.1, 90
for t_cd in [0, 15, 45, 75, 90]:
    print(t_cd, round(s0_cd / 2 * (1 + math.cos(math.pi * t_cd / tau_cd)), 6));
#> 0 0.1
#> 15 0.093301
#> 45 0.05
#> 75 0.006699
#> 90 0.0
```

The step length is 0.1 at the start, half of that at $t = 45$, and zero at $t = 90$. It changes slowly near both ends and fastest in the middle.

Remark (Restarting the cosine schedule)

Remark 0.4 (Restarting the cosine schedule). Restarting the schedule periodically is a common variant of cosine decay [8]: jump the step length back to $s^{(0)}$ and decay again. One motivation is that the jump can move the iterates out of one basin of attraction and into a better one.

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